Mod 3.4

12 October 2026 08:07

$N = 3$

$\vec{x_1} = (2, 2) \quad \vec{x_2} = (4, 5)$

$\vec{x_3} = (7, 4) \quad y_1 = -1 \quad y_2 = +1 \quad y_3 = +1$

x_1	x_2	Class
2	2	−1
4	5	+1
7	4	+1

$\vec{a} = (a_1, a_2, a_3)$

$$= \sum_{i=1}^{N} a_i y_i = -a_1 + a_2 + a_3$$

$$\phi(\vec{a}) = \sum_{i=1}^{N} a_i - \frac{1}{2} \sum_{i,j=1}^{N} a_i a_j y_i y_j (\vec{x_i} \cdot \vec{x_j})$$

$(\vec{x_1} \cdot \vec{x_1}) = 8$ $\quad (2,2)(2,2) = 4+4 = 8$

$(\vec{x_1} \cdot \vec{x_2}) = 18$ $\quad (2,2)(4,5) = 8+10 = 18$

$(\vec{x_1} \cdot \vec{x_3}) = 22$ $\quad (\vec{x_2}, \vec{x_2}) = 41$ $\quad (\vec{x_2}, \vec{x_3}) = 48$

$(\vec{x_2} \cdot \vec{x_1}) = 18$ $\quad (\vec{x_3}, \vec{x_1}) = 22$ $\quad (\vec{x_3}, \vec{x_2}) = 48$

$(\vec{x_3}, \vec{x_3}) = 65$

$$\phi(\vec{a}) = (a_1 + a_2 + a_3) - \frac{1}{2}\left[(a_1 a_1 (-1)(-1)(8)) + \{a_1 a_2 (-1)(+1)(18)\}\right.$$

$$\left. + \{a_1 a_3 (-1)(+1)(22)\} + \{a_2 a_1 (+1)(-1)(18)\} + \{a_2 a_2 (+1)(+1)(41)\}\right]$$

$$+\{a_2a_3(+1)(+1)(48)\}+\{a_3a_1(+1)(-1)(22)\}+\{a_3a_2(+1)(+1)(48)\}$$
$$+\{a_3a_3(+1)(+1)(65)\}]$$

$$=(a_1+a_2+a_3)-\frac{1}{2}[8a_1^2-18a_1a_2-22a_1a_3-18a_2a_1+41a_2^2$$
$$+48a_2a_3-22a_3a_1+48a_3a_2+65a_3^2]$$

$$=(a_1+a_2+a_3)-\frac{1}{2}[8a_1^2-36a_1a_2-44a_1a_3+41a_2^2$$
$$+96a_3a_2+65a_3^2]$$

$$=2(a_2+a_3)-\frac{1}{2}[13a_2^2+32a_2a_3+25a_3^2]$$

$$\left[\because\ a_1=a_2+a_3\right]$$

$$\frac{\partial\phi}{\partial a_2}=2-13a_2-16a_3$$

$$\frac{\partial\phi}{\partial a_3}=2-16a_2-25a_3$$

$$a_2=\frac{26}{121}$$

$$a_3=-\frac{6}{121}$$

$$a_1=a_2+a_3=\frac{26}{121}-\frac{6}{121}=\frac{26}{121}$$

$$\vec{w}=\sum_{i=1}^{N}a_i\,y_i\,\vec{x_i}$$

$$= \frac{20}{121}(-1)(2,2) + \frac{26}{121}(+1)(4,5) - \frac{6}{121}(+1)(7,4)$$

$$\vec{w}_x = \frac{20}{121}(-1)(2) + \frac{26}{121}(+1)(4) - \frac{6}{121}(+1)(7) = \boxed{\frac{2}{11}}$$

$$\vec{w}_y = \frac{20}{121}(-1)(2) + \frac{26}{121}(+1)(5) - \frac{6}{121}(+1)(4) = \boxed{\frac{6}{11}}$$

$$b = \frac{1}{2}\left(\min_{y=+1}(\vec{w}\cdot\vec{x_i}) + \max_{y=-1}(\vec{w}\cdot\vec{x_i})\right)$$

$$\vec{w}\cdot\vec{x_1} = \left(\frac{2}{11}\times 2 + \frac{6}{11}\times 2\right) = \frac{16}{11}$$

$$\vec{w}\cdot\vec{x_2} = \left(\frac{2}{11}\times 4 + \frac{6}{11}\times 5\right) = \frac{38}{11}$$

$$\vec{w}\cdot\vec{x_3} = \left(\frac{2}{11}\times 7 + \frac{6}{11}\times 4\right) = \frac{38}{11}$$

$$= \frac{1}{2}\left(\min\left\{\frac{38}{11}, \frac{38}{11}\right\} + \max\left\{\frac{16}{11}\right\}\right)$$

$$= \frac{1}{2}\left(\frac{38}{11} + \frac{16}{11}\right) = \boxed{\frac{27}{11}}$$

$$f(\vec{x}) = \vec{w}\cdot\vec{x} - b$$

$$\boxed{= \frac{2}{11}x_1 + \frac{6}{11}x_2 - \frac{27}{11}}$$

Multiply by α_1 & divide by 2

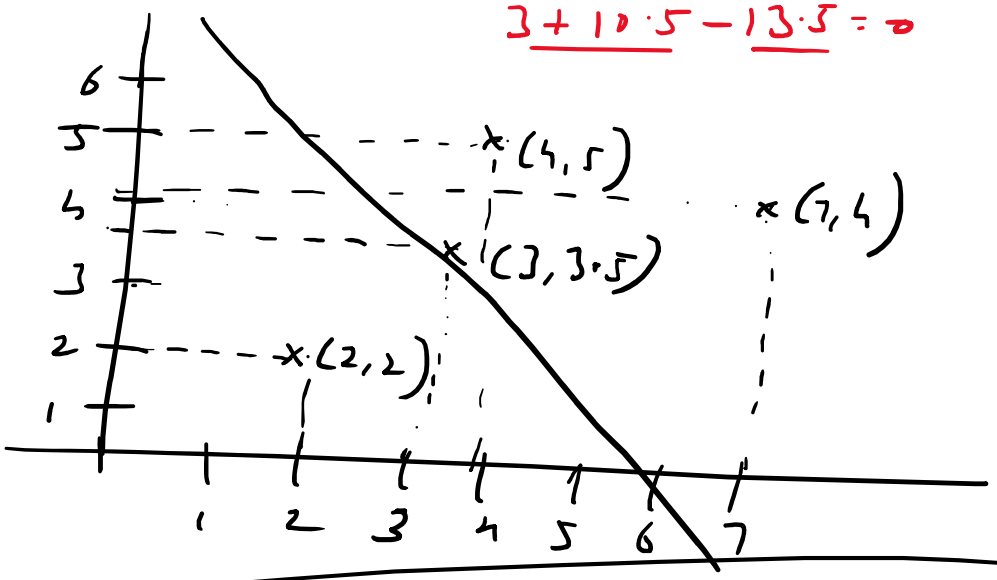
$$= \alpha_1 + 3\alpha_2 - \frac{27}{2} = 0$$

[a_3 is not support vector as it is −ve]

$3 + 10.5 - 13.5 = 0$

$x_1 = \frac{2+4}{2} = 3$

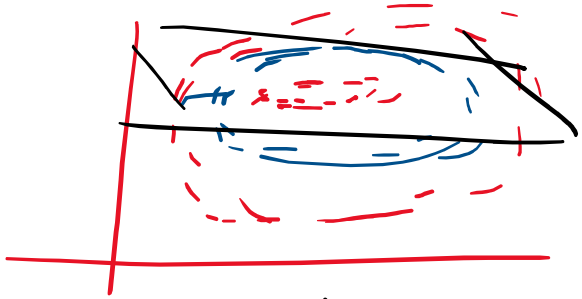
$x_2 = \frac{2+5}{2} = 3.5$

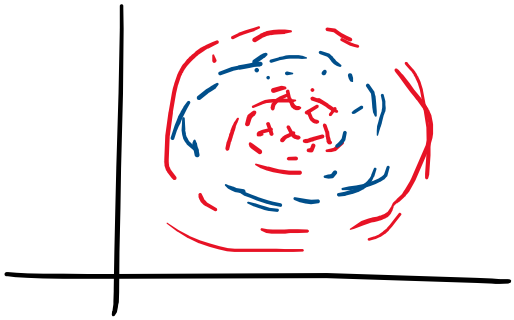


=> Non-linear data handling using SVM :

Kernelization Trick :

1) Polynomial Kernel





2) Radial Basis Fitting

ii) RBF kernel ~ Radial Basis Fitting